

02st-10-2026

International Cybersecurity Of Digital Forensic Academy (ICDFA)

PROGRAMME: Advance Critical Infrastructure Security (ACIS)

Prepared by: Aminu Ibrahim Jalo
Course: ACICP301-INTRO TO CRITICAL INFRASTRUCTURE
FOR IT LEARNERS

Title: Introduction to Human Machine Interface (HMI)

Assignment: Week 1-Lab 2 Report
C11/26/ACIS/17359

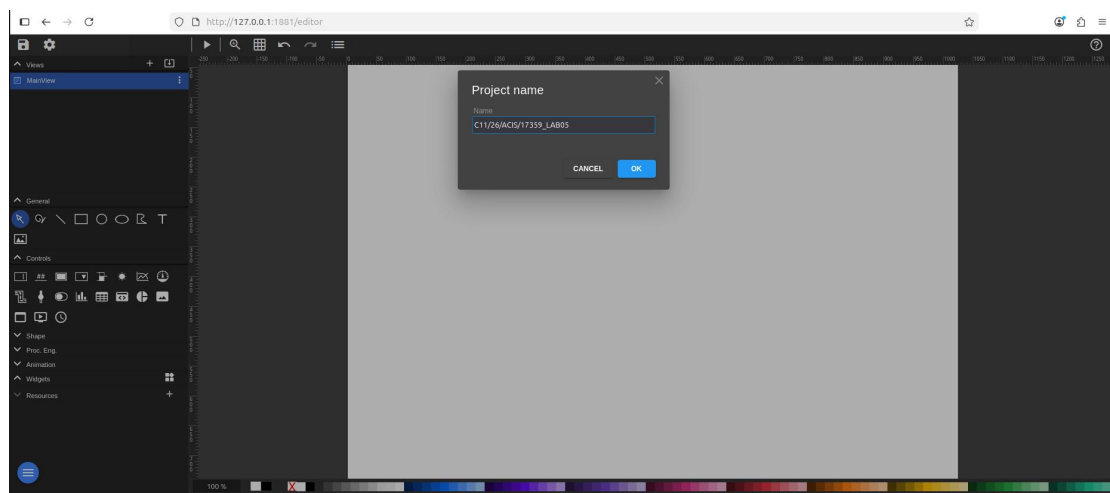
Instructor : Aminu Idris

Student Name	Aminu Ibrahim Jalo
Registration Number	C11/26/ACIS/17359
Signature	
Date	02 st -10-2026

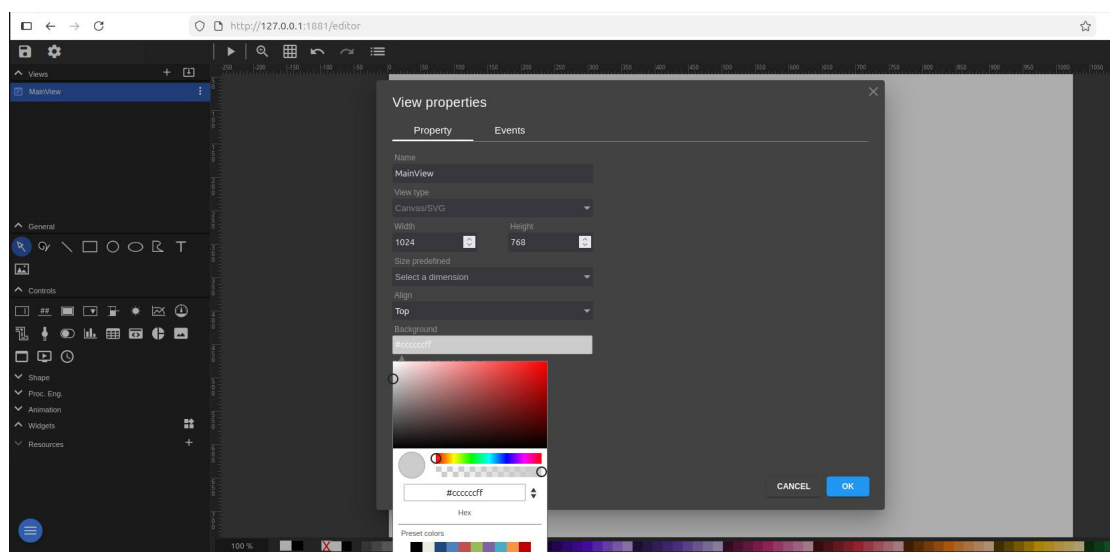
Procedure

Step 1. Start the PLC and SCADA VMs in the isolated laboratory network. Start FUXA on the SCADA VM and open <http://127.0.0.1:1881/editor>.

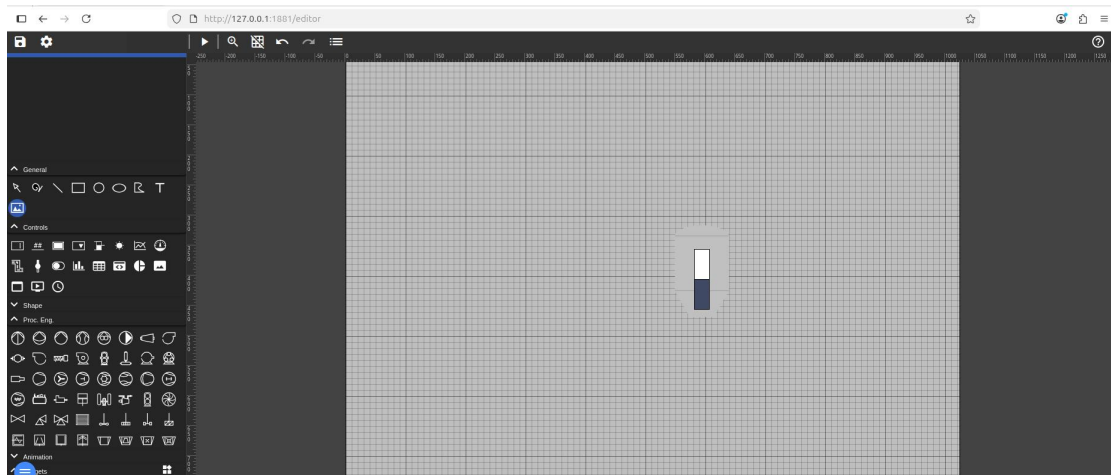
Step 2. Create a New Project named with your registration number and LAB05, for example 2026XXXX_LAB05.



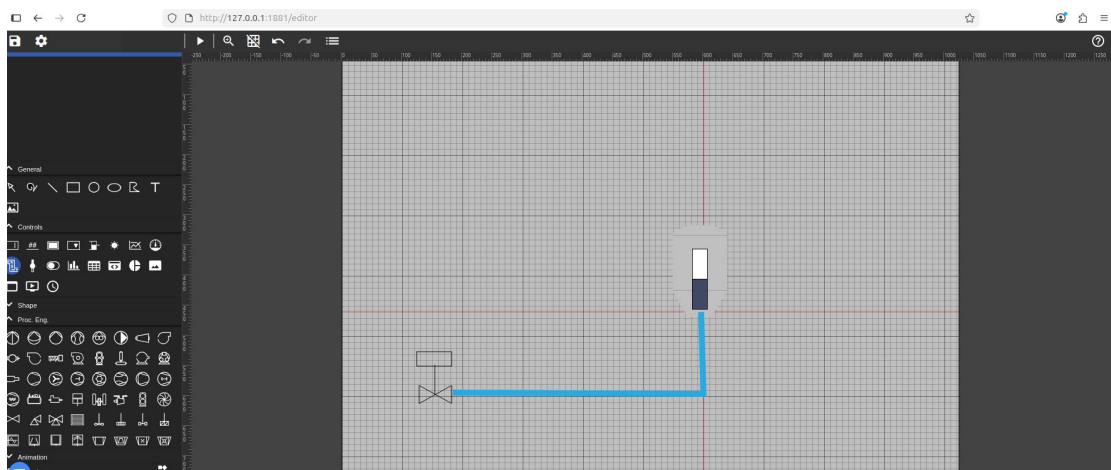
Step 3. Set a neutral gray background and enable the grid while arranging process graphics.



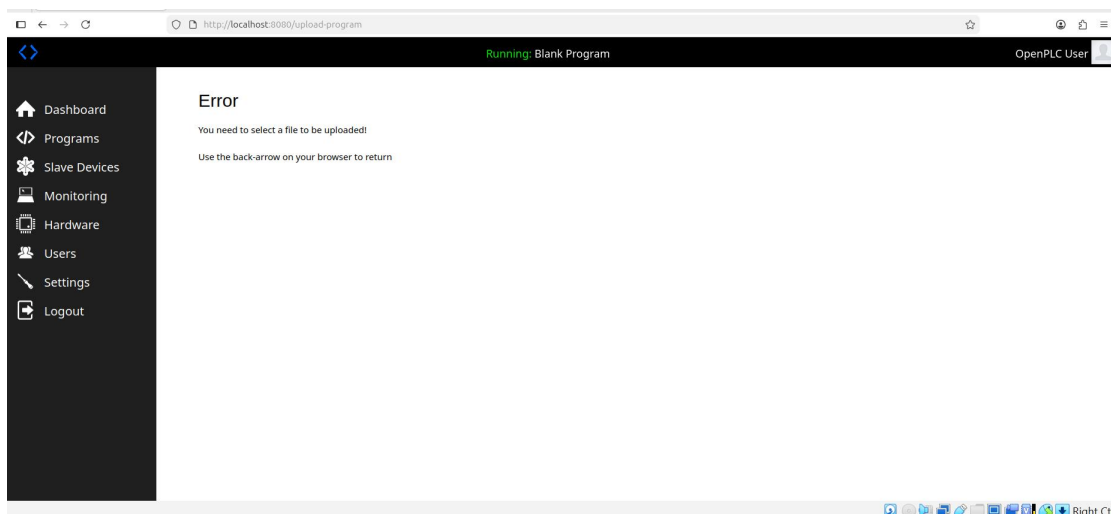
Step 4. From the Process Engineering objects, place a tank graphic. Add a bar gauge over/inside the tank and set its engineering range to 0–10 ft.



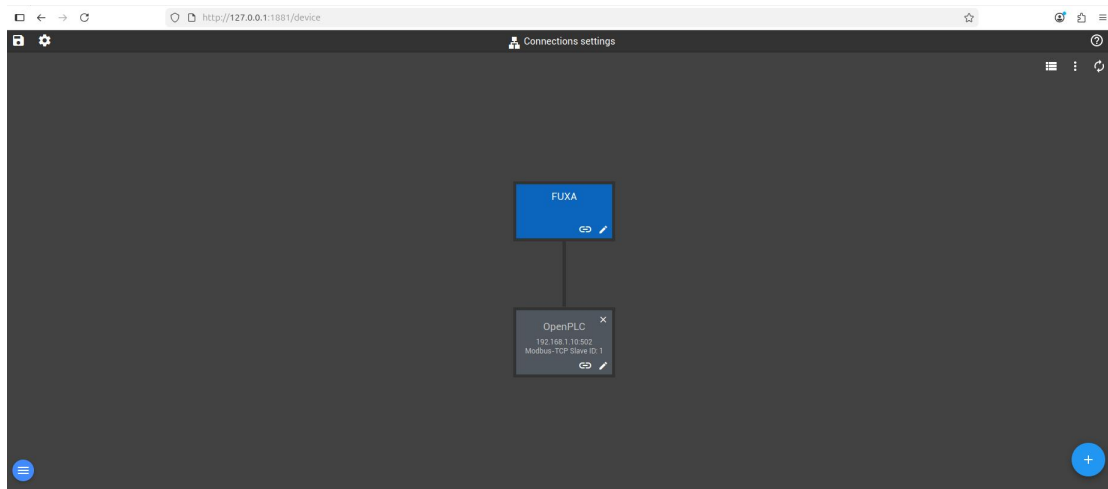
Step 5. Build a simple motor-actuated fill valve using the available valve/shape objects. Add a pipe from the valve to the tank. Keep the normal-state design visually simple.



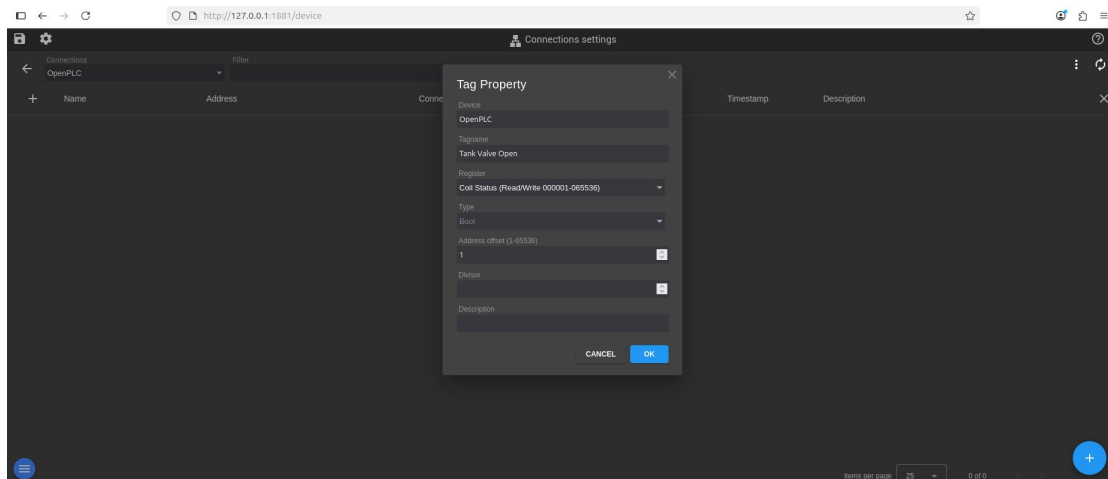
Step 6. On the PLC VM, load/start the provided TankFillSim.st program in OpenPLC Runtime and confirm the status is Running.



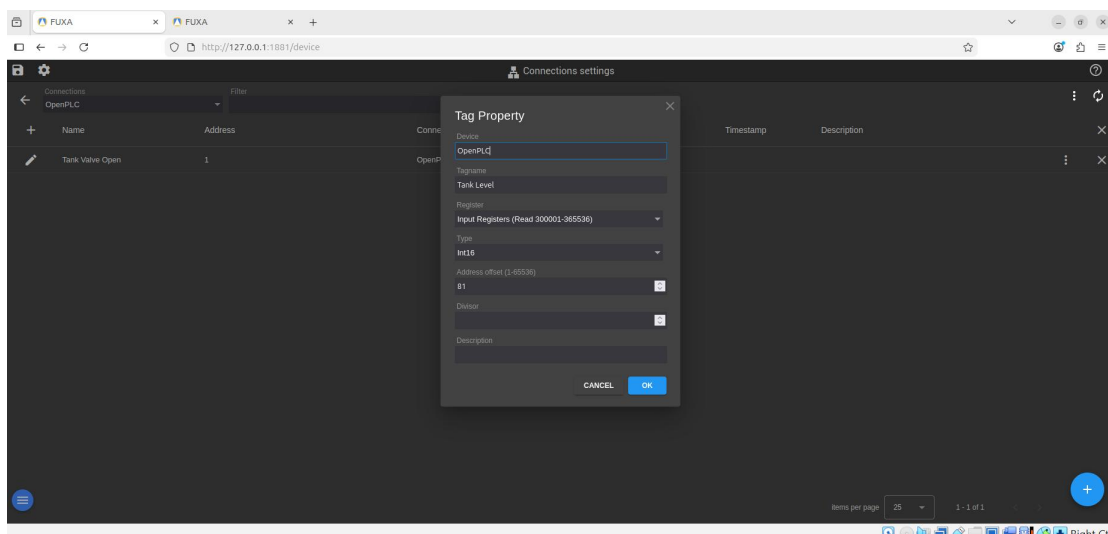
Step 7. In FUXA, create or edit the Modbus TCP connection to the PLC at 192.168.1.10:502.



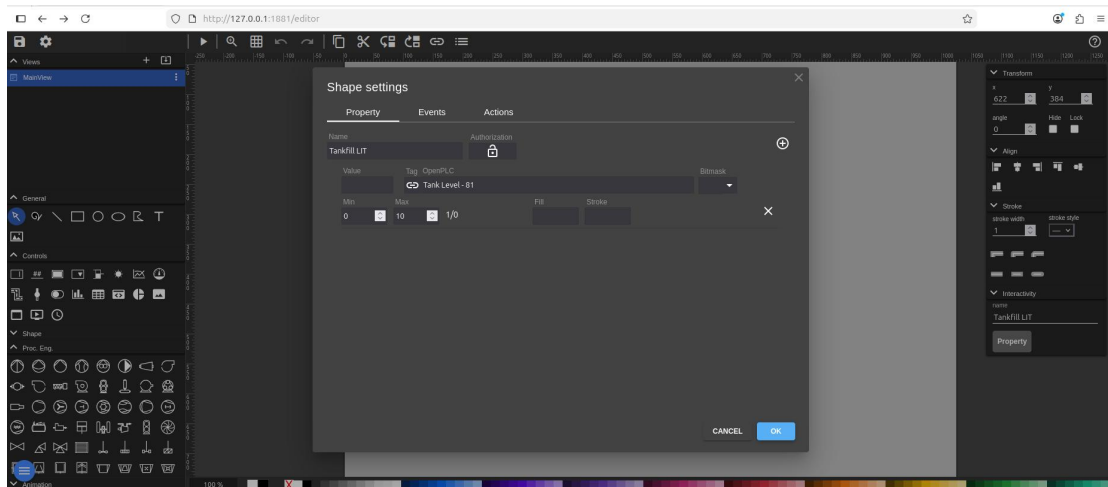
Step 8. Create/verify the Tank Level tag mapped to the OpenPLC input-register location used by the simulation (%IW80 / Input Register 80 in the teaching image).



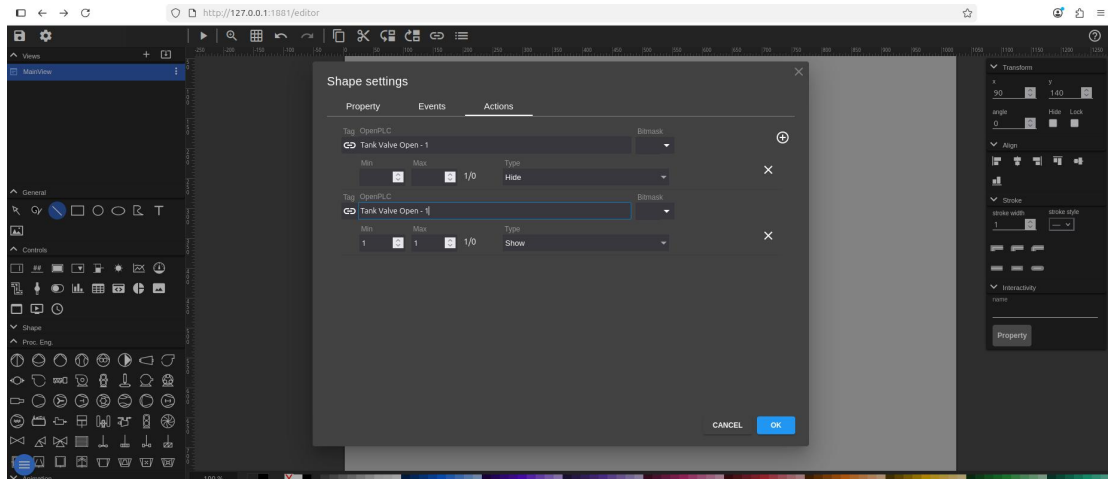
Step 9. Create/verify the Tank Valve Open tag mapped to the OpenPLC discrete output used by the simulation (%QX0.0 / Coil 0 in the teaching image).



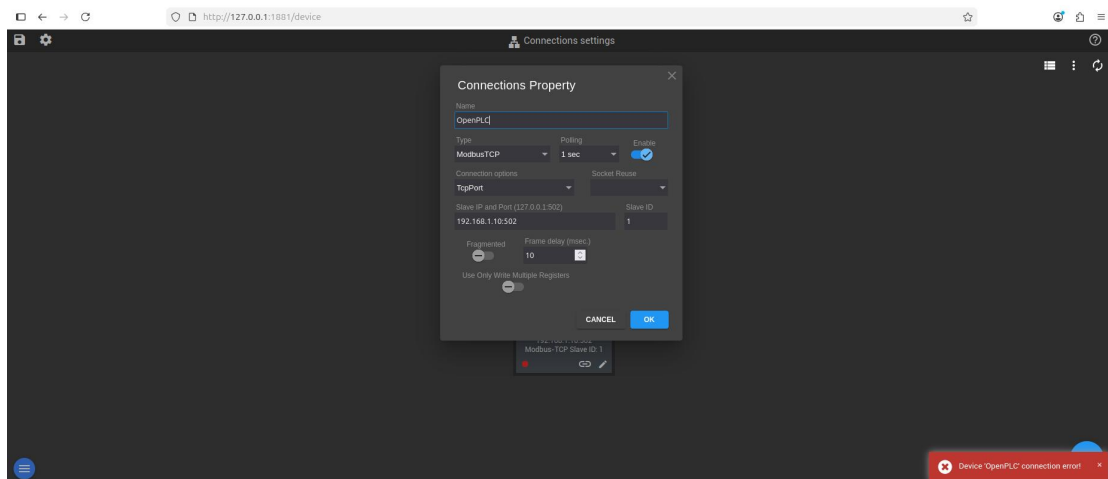
Step 10. Bind the bar gauge to the Tank Level tag. Configure Min = 0 and Max = 10.

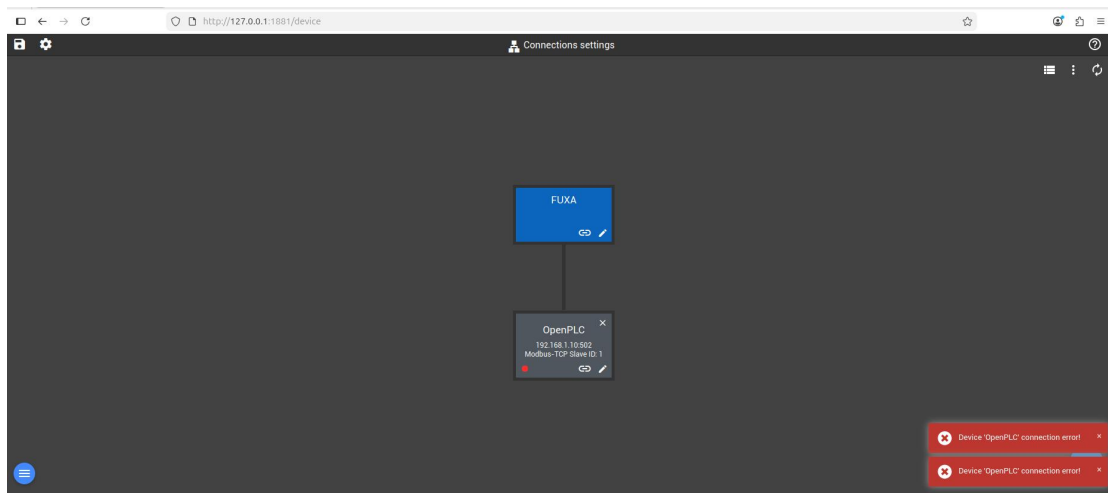


Step 11. Bind the valve-state graphic to Tank Valve Open. Configure the normal/closed representation for value 0 and the open representation for value 1.



Step 12. Use a restrained high-performance style: neutral normal states, clear labels, and attention-drawing color only where it communicates a meaningful abnormal/alarm condition.





Step 13. Enable the PLC/FUXA connection and run the HMI. Observe the tank filling/draining and confirm the valve-state graphic changes with the process.

Step 14. Save/export the FUXA project and capture evidence of both the design view and running HMI.

HMI design rationale

The HMI was designed using high-performance visualization principles to support operator situational awareness. A neutral gray background provides a consistent visual environment, while the tank-level gauge displays the process measurement within a defined range of 0-10 ft. The valve-state graphic communicates whether the fill valve is open or closed, and the piping layout helps operators understand the process flow. Restrained use of color reduces distraction, while clear labels and meaningful state indications help operators identify changes in the simulated tank-fill process. The HMI is connected to the OpenPLC simulation through Modbus TCP to display process information from the configured tags.

Knowledge Check

1. What is a P&ID and how does it influence HMI layout?

Response: A P&ID (Piping and Instrumentation Diagram) is a technical drawing that represents process equipment, piping, valves, instruments and their connections. It guides HMI layout by showing the relationships between process components, helping operators understand the process flow, identify equipment and monitor process conditions.

2. Why do high-performance HMIs avoid unnecessary animation and excessive color?

Response: High-performance HMIs avoid unnecessary animation and excessive color to reduce operator distraction, cognitive workload and alarm fatigue. Neutral colors are used for normal operating conditions, while distinctive colors are reserved for important abnormal conditions and alarms. This helps operators recognize significant changes quickly and make informed decisions.

3. How does the Tank Level tag differ from the Tank Valve Open tag in data type and behavior?

Response: The Tank Level tag represents a numerical process measurement, such as the tank level in feet, and is displayed using a gauge or numerical indicator with a

range of 0–10 ft. The Tank Valve Open tag represents a discrete Boolean state, typically 0 for closed and 1 for open, and is displayed using a valve graphic that changes according to the state.

4. What could happen operationally if an HMI displays stale or incorrect PLC data?

Response: Stale or incorrect PLC data can mislead operators about the actual process conditions. Operators may make inappropriate control decisions, fail to recognize abnormal tank levels, incorrectly assume a valve is open or closed, or respond too late to a developing fault. These issues can cause process interruptions, equipment damage, safety incidents or production losses. Data-quality indicators, communication-status monitoring, alarms and validation checks can help reduce these risks.